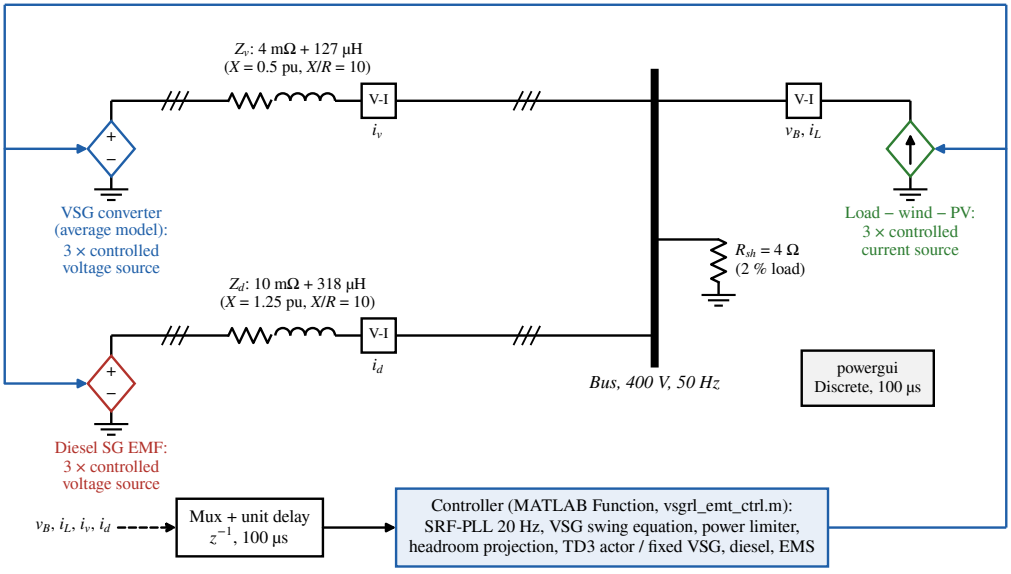
source references: VSG and diesel phase voltages, injected phase currents
Z_v: 4 mΩ + 127 µH
(X = 0.5 pu, X/R = 10)
V-I
i_v
V-I
v_B, i_L
VSG converter
(average model):
3 × controlled
voltage source
Load – wind – PV:
3 × controlled
current source
R_{sh} = 4 Ω
(2 % load)
Z_d: 10 mΩ + 318 µH
(X = 1.25 pu, X/R = 10)
V-I
i_d
Bus, 400 V, 50 Hz
powergui
Discrete, 100 µs
Diesel SG EMF:
3 × controlled
voltage source
v_B, i_L, i_v, i_d
Mux + unit delay
z^{-1}, 100 µs
Controller (MATLAB Function, vsgrl_emt_ctrl.m):
SRF-PLL 20 Hz, VSG swing equation, power limiter,
headroom projection, TD3 actor / fixed VSG, diesel, EMS